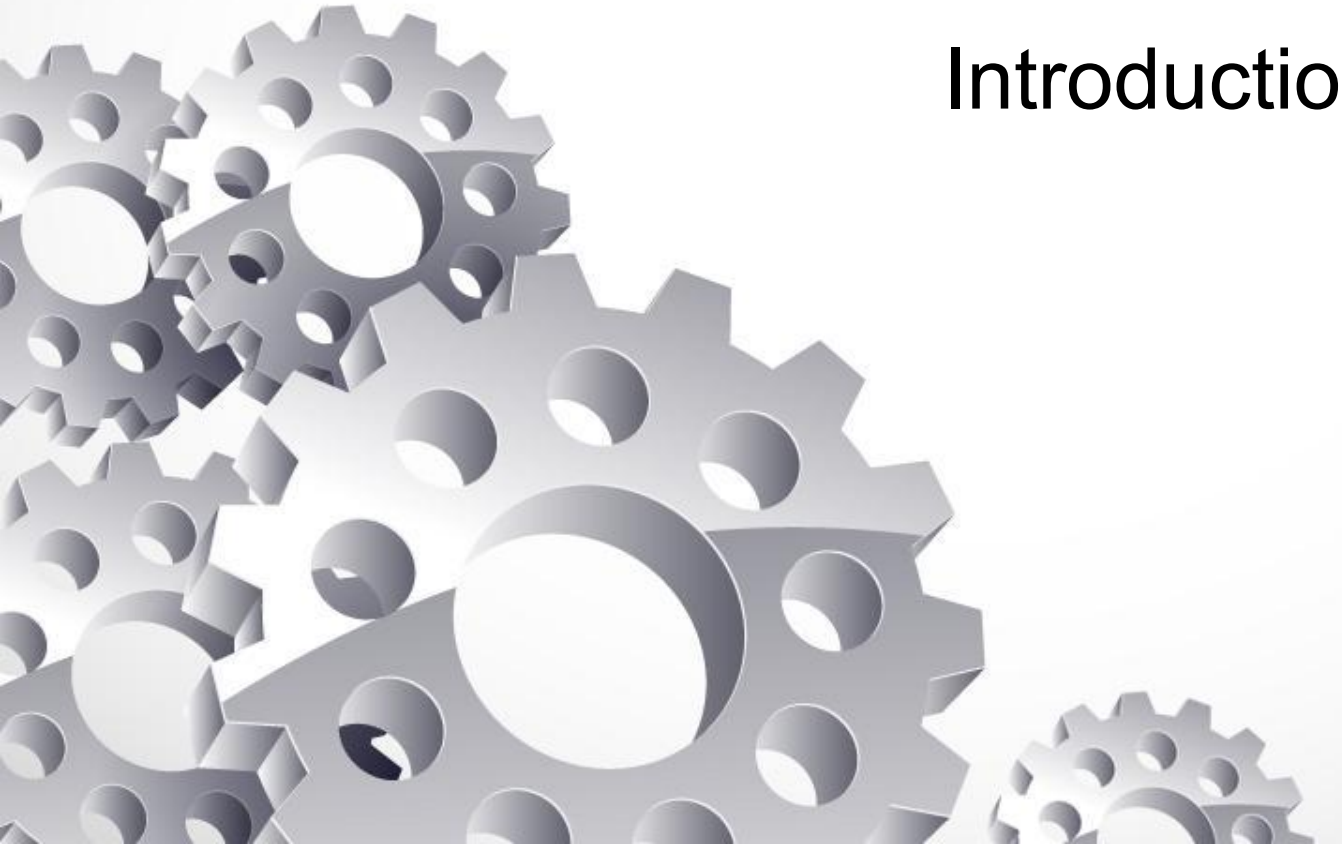
Web Technology SCSJ 3483



Introduction to jQuery

What is jQuery?



- a lightweight, "write less, do more", JavaScript library.
- The purpose is to make it much easier to use JavaScript.
- Simplifies a lot of the complicated things from JavaScript, like AJAX calls and DOM manipulation.

jQuery Javascript Library



- The jQuery library contains the following features:
  - HTML/DOM manipulation
  - CSS manipulation
  - HTML event methods
  - Effects and animations
  - AJAX
  - Utilities

Using jQuery



- several ways to start using jQuery on your web site (refer code/boilerplate):
  - Download
    - jquery.com
  - Using cdn

Downloading jQuery

- There are two versions of jQuery available for downloading:
- Production version - this is for your live website because it has been minified and compressed (*.min.js)
- Development version - this is for testing and development (uncompressed and readable code)

Downloading jQuery



- Both versions can be downloaded from jQuery.com.
- Refer in html – put before closing body tag
- <script src="js/jquery-3.2.1.min.js"></script>

Alternatives to Downloading



- If you don't want to download and host jQuery yourself, you can include it from a CDN (Content Delivery Network).
- Both Google and Microsoft host jQuery.

Google CDN



- <script
  - src="https://ajax.googleapis.com/ajax/libs/jquery/3.2.1/jquery.min.js">
- </script>

Hello World



- Sample code (code/sample)
  - hello1.html – local scripts
  - hello2.html – Google CDN

jQuery Syntax



- The jQuery syntax is tailor made for **selecting** HTML elements and performing some **action** on the element(s).
- Basic syntax is: **$(*selector*).*action*()**
  - A **$** sign to define/access jQuery
  - A (***selector***) to "query (or find)" HTML elements
  - A jQuery ***action***() to be performed on the element(s)

jQuery Syntax



- Examples:
  - $(this).hide() - hides the current element.
  - $("p").hide() - hides all <p> elements.
  - $(".test").hide() - hides all elements with class="test".
  - $("#test").hide() - hides the element with id="test".

The Document Ready Event



- to prevent any jQuery code from running before the document is finished loading (is ready).
  - No problem if follow common practice on where to put javascript code
- $(document).ready(function(){

  // jQuery methods go here...

  });

The Document Ready Event



- Good practice to wait for the document to be fully loaded and ready before working with it.
- Some examples of actions that can fail if methods are run before the document is fully loaded:
  - Trying to hide an element that is not created yet
  - Trying to get the size of an image that is not loaded yet

The Document Ready Event



- **Shortcut**:
- $(function(){

```
    // jQuery methods go here...

});
```

jQuery Selectors



- jQuery selectors allow selection and manipulation of HTML element(s).
- Enable to find elements based on their id, classes, types, attributes, values of attributes and much more.
- It's based on the existing CSS Selectors, and in addition, it has some own custom selectors.
- All type of selectors in jQuery, start with the dollar sign and parentheses: $().

Example – element selector

- select all <p> elements on a page like this:
  - $("p")
- When a user clicks on a button, all <p> elements will be hidden:
  -
```
$(document).ready(function(){
  $("button").click(function(){
    $("p").hide();
  });
});
```

Example – #id selector



- #id selector uses the id attribute of an HTML tag to find the specific element.
- id should be unique within a page, so the #id selector is use to find a single, unique element
  - $(document).ready(function(){
    $("button").click(function(){
    $("#test").hide();
    });
    });

Example – .class Selector

- finds elements with a specific class – by writing a period character, followed by the name of the class - $(".test")
  - $(document).ready(function(){
    $("button").click(function(){
      $(".test").hide();
    });
  });

Check sample code



- element, id and class

More Examples of jQuery Selectors

Syntax	Description
$("*")	Selects all elements
$(this)	Selects the current HTML element
$("p.intro")	Selects all <p> elements with class="intro"
$("p:first")	Selects the first <p> element
$("ul li:first")	Selects the first element of the first
$("ul li:first-child")	Selects the first element of every
$("[href]")	Selects all elements with an href attribute
$("a[target='_blank']")	Selects all <a> elements with a target attribute value equal to "_blank"
$("a[target!='_blank']")	Selects all <a> elements with a target attribute value NOT equal to "_blank"
$(":button")	Selects all <button> elements and <input> elements of type="button"
$("tr:even")	Selects all even <tr> elements
$("tr:odd")	Selects all odd <tr> elements

jQuery DOM



- Sample code – dom folder
- Retieving DOM info for current node

- Hello World

- var current = $('#para'); //get the node using ID
- var currentNode = $(current)[0]; //get the native DOM (element ZERO)
- console.log("Node type: " + currentNode.nodeType); // 1
- console.log("Node name: " + currentNode.nodeName); // P
- console.log("Node ID: " + currentNode.id); // para

jQuery DOM



- Get parent
  - <body>
    - Hello World
  - </body>

- var parent = $('#para').parent();
- console.log($(parent)[0].nodeName) // body

jQuery DOM



- Get children
  - <body>
    - Hello < i>World</i>
  - </body>

- var children = $('#para').children(); //this will ignore textNode
- var children = $('#para').contents(); //include textNode

jQuery DOM



- Traversing

```
children.each(function(){
    var nativeNode = $(this)[0]; //get the native DOM element
    var node = $(this); //as normal jquery node
});
```

jQuery DOM



- Get first child

- var first = $('#para').children().eq(0);
- var nativeNode = $(first)[0]; //get the native DOM element

Important jQuery DOM Functions



- $().parseHTML('); //createElement
- $().html(); //get or set element
- $().val(); //get text value for form element
- $().append(); //append text or html
- $().attr(); //get or set attribute

Important jQuery DOM Functions



- code\sample folder
  - create.html
  - prepend.html. append.html
  - getset.html
  - attr1.html, attr2.html
- code\table
  - Dynamic table

jQuery Events

Event Method	Description
$(selector).click(function)	Invokes a function when the selected elements are clicked
$(selector).dblclick(function)	Invokes a function when the selected elements are double-clicked
$(selector).focus(function)	Invokes a function when the selected elements receive the focus
$(selector).mouseover(function)	Invokes a function when the mouse is over the selected elements
$(selector).keypress(function)	Invokes a function when a key is pressed inside the selected elements

For a full jQuery event reference, please see jQuery Events Reference

Manipulating CSS

CSS Properties	Description
$(selector).css(propertyName)	Get the style property value of the first selected element
$(selector).css(propertyName,value)	Set the value of one style property for selected elements
$(selector).css({properties})	Set multiple style properties for selected elements
$(selector).addClass(class)	Apply style *class* to the selected elements

For a full jQuery CSS reference, please see jQuery CSS Methods Reference
For more on the css function, see http://api.jquery.com/css/

jQuery Effects

Function	Description
$(selector).hide()	Hide selected elements
$(selector).show()	Show selected elements
$(selector).toggle()	Toggle (between hide and show) selected elements
$(selector).slideDown()	Slide-down (show) selected elements
$(selector).slideUp()	Slide-up (hide) selected elements
$(selector).slideToggle()	Toggle slide-up and slide-down of selected elements
$(selector).fadeIn()	Fade in selected elements
$(selector).fadeOut()	Fade out selected elements
$(selector).fadeTo()	Fade out selected elements to a given opacity
$(selector).fadeToggle()	Toggle between fade in and fade out

Events, css, effect, form example



- code\sample
  - Css and fadein fadeout
    - various.html
- code\Zebra
- code\Focusblur
- code\tablerow
- code\jqform
  - client side form operation – listbox, radio, checkbox
- code\forminfo
  - Form field information
- code\formpost
  - post method with no refresh

All In – Table Operations



- tableop (dynamic table and crud)

All In – Table Operations

- Set the table scaffolding

```
<table border="1" id="tbl1">
  <thead>
    <tr>
      <th scope="col">Index</th>
      <th scope="col">Title</th>
      <th scope="col">Author</th>
      <th scope="col">Delete</th>
    </tr>
  </thead>
  <tbody>
  </tbody>
</table>
```

Dynamic TR – TD, DATA attr



- HTML5 data
  - data-* attributes is used to store custom data private to the page or application.
  - data-* attributes gives us the ability to embed custom data attributes on all HTML elements (DOM).
  - Use the first td (index = 0)
  - var tdIndex = $.parseHTML('<td></td>');
  - Set value
    - $(tdIndex).data("isbn", isbn);
  - Get value
    - var isbn = $(firstTD).data("isbn");

Dynamic TR – TD link and image link



- TD link – 2nd td (index 1)
  - var tdTitle = $.parseHTML('<td></td>');
  - $(tdTitle).html("" + title + "");
- TD image link – 4th td (index 3)
  - var tdDelete = $.parseHTML('<td></td>');
  - $(tdDelete).html(" <img src=/img/del.png' height='20'> ");

Append TR to tbody



```
addDataToTableRow(1, "isbn123", "The Way of Kings", "Brandon Sanderson", "TOR", "13/02/2010", "9.99");
addDataToTableRow(2, "isbn456", "Kingkiller Chronicles", "Patrick Rothfuss", "Gollantz", "13/02/2012", "12.99");
```

```
function addDataToTableRow(index, isbn, title, author, publisher, publishdate, price) {
  var row = $.parseHTML('<tr></tr>');

  var tdIndex = $.parseHTML('<td></td>');
  $(tdIndex).html(index);
  $(tdIndex).data("isbn", isbn);
  $(tdIndex).data("title", title);
  $(tdIndex).data("author", author);
  $(tdIndex).data("publisher", publisher);
  $(tdIndex).data("publishdate", publishdate);
  $(tdIndex).data("price", price);

  var tdTitle = $.parseHTML('<td></td>');
  $(tdTitle).html("<a href='#'>" + title + "</a>");

  var tdAuthor = $.parseHTML('<td></td>');
  $(tdAuthor).html(author);

  var tdDelete = $.parseHTML('<td></td>');
  $(tdDelete).html("<div align='center'><a href='#'><img src='http://gmm.fsksm.utm.my/~rosely/scj4483/jquery/images/del.png'

  //add col to row
  $(row).append(tdIndex);
  $(row).append(tdTitle);
  $(row).append(tdAuthor);
  $(row).append(tdDelete);

  //add row to table
  $("#tbl1 tbody").append(row);
}
```

- Finding clicked TD
- Then find parent TR
- Then find first TD for DATA retrieval

```
$("#tbl1").on("click", "td", function(e) {

    var tdIndex = $(this).index();
    console.log(tdIndex);
    console.log($(this).html()) //td content

    var parentTR = $(this).parent();
    var firstTD = $(parentTR).children().eq(0);
```

```
if (tdIndex == 1) //clicking title link - popup view/edit all data
{
  //isbn
  //data attribute in 1st td
  var isbn = $(firstTD).data("isbn");

  //title
  //data attribute in 1st td
  var title = $(firstTD).data("title");

  //author
  //data attribute in 1st td
  var author = $(firstTD).data("author");

  //publisher
  //data attribute in 1st td
  var publisher = $(firstTD).data("publisher");

  //publisher
  //data attribute in 1st td
  var publishdate = $(firstTD).data("publishdate");

  //publishdate
  //data attribute in 1st td
  var publishdate = $(firstTD).data("publishdate");

  //price
  //data attribute in 1st td
  var price = $(firstTD).data("price");

  alert(isbn + "\n" + title + "\n" + author + "\n" + publisher + "\n" + publishdate + "\n" + price);
}
```

- When updating value in TD, DATA attribute must also be updated

```
if (tdIndex == 2) //clicking author td
{
  var author = $(this).html();

  var name = window.prompt("Please enter new name", author);

  if (name !== null)
  {
    $(this).html(name);

    //DATA attribute also must be updated
    $(firstTD).data("author", name);
  }
}
```

- Mouseleave and mouseenter for making sure the user know that there something can be done with the TD value

- Finding TR using parent traversal based on IMG click

- Deleting element using fadeOut effect and callback for DOM element removal
  - Asyn problem in browser operation

```
$("#tbl1").on("mouseenter", "td", function(e) {
  var tdIndex = $(this).index();
  if (tdIndex == 2) {
    $(this).css('cursor', 'pointer');
    $(this).css('font-weight', 'bold');
  }
}).on("mouseleave", "td", function() {
  var tdIndex = $(this).index();
  if (tdIndex == 2) {
    $(this).css('cursor', 'default');
    $(this).css('font-weight', 'normal');
  }
});

$("#tbl1").on("click", "img", function() {
  //                      A       DIV      TD       TR
  var parentTR = $(this).parent().parent().parent().parent();
  //alert(parentTR.html());

  var answer = confirm("Are you sure you want to delete the record?")

  if (answer) {
    $(parentTR).fadeOut("slow", "swing", function(e) {
      $(parentTR).remove();
    });
  }
});
```